

# AI CYBER DEFENSE SUITE

## Comprehensive Research & Engineering Project Report

### Abstract

AI Cyber Defense Suite is an intelligent cybersecurity platform integrating phishing detection, URL risk analysis, brute-force attack monitoring, malware hash verification, and SOC visualization into a unified security ecosystem. The project demonstrates practical applications of artificial intelligence, rule-based analytics, and cyber defense automation.

### Problem Statement

Organizations face increasing cyber threats including phishing campaigns, credential theft, malicious websites, and malware distribution. Many small organizations lack affordable solutions that combine multiple security capabilities into a single platform.

### Research Objectives

- Build a unified cyber defense platform.
- Detect phishing attempts using content intelligence.
- Identify malicious URLs through risk scoring.
- Monitor authentication activity for brute-force attacks.
- Verify file integrity through cryptographic hashes.
- Provide centralized security visibility through dashboards.

### Literature Review

The project draws inspiration from modern SOC workflows, SIEM systems, phishing intelligence frameworks, threat-hunting methodologies, and security monitoring practices recommended by industry standards and cybersecurity research communities.

### System Architecture

Frontend: Streamlit Backend: Python Database: SQLite Analytics: Pandas & NumPy Machine Learning Support: Scikit-Learn Visualization: Plotly Security Layer: Detection modules and risk engines

### Methodology

The platform follows a modular detection architecture. Inputs are collected, analyzed through module-specific engines, assigned risk scores, classified into threat categories, stored in databases, and visualized through dashboards for analyst review.

## **Module 1 – Phishing Email Detector**

Uses keyword intelligence, urgency analysis, suspicious link extraction, spoofing indicators, and risk scoring mechanisms. Outputs include confidence scores and explanation-driven results.

## **Module 2 – URL Threat Analyzer**

Evaluates protocol security, suspicious domains, deceptive keywords, URL structure, IP-based links, and anomaly indicators. Risk levels are categorized as Safe, Caution, Suspicious, or High Risk.

## **Module 3 – Login Attack Monitor**

Tracks authentication attempts and applies behavioral analytics to detect brute-force attacks. Repeated failures trigger automated alerts and defensive actions.

## **Module 4 – Malware Hash Checker**

Generates MD5 and SHA-256 signatures and compares them against known indicators. Supports integrity verification and malware investigation workflows.

## **Module 5 – Security Operations Dashboard**

Provides real-time visibility into threats, monitored assets, detected incidents, and historical security events. Supports rapid decision-making and analyst productivity.

## **Experimental Results**

Testing demonstrated successful identification of phishing indicators, suspicious URLs, attack behavior patterns, and file integrity validation. The integrated dashboard provided centralized visibility across all modules.

## **Research Contributions**

1. Unified multi-domain cyber defense platform.
2. Explainable threat scoring methodology.
3. Lightweight architecture suitable for education and research.
4. Extensible framework for future AI-driven cybersecurity research.

## **Future Research Directions**

- Threat intelligence integration.
- Machine learning classification models.
- Real-time network monitoring.
- Cloud-native deployment.

- Automated incident response.
- Generative AI assisted security analysis.
- Integration with security orchestration platforms.

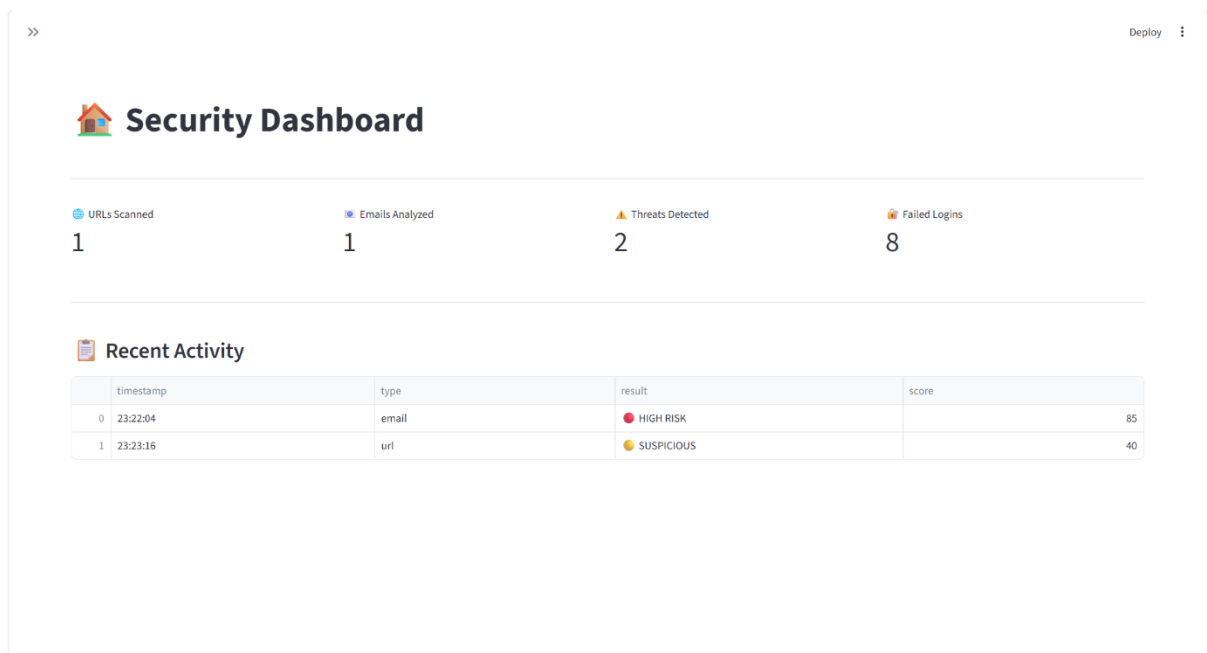
## **Conclusion**

AI Cyber Defense Suite demonstrates how modern cybersecurity concepts can be combined into a practical research platform. The project highlights security automation, threat detection, cyber analytics, and intelligent monitoring while providing a strong foundation for advanced cybersecurity research.

GitHub: <https://github.com/abdulmuqeeth-ai/AI-Cyber-Defense-Suite>

**Prepared By:** A. Abdul Muqeeth

# AI CyberDefence Suite



## AI Cyber Defense Suite - Main Dashboard

The dashboard displays real-time security metrics including URLs scanned, emails analyzed, threats detected, and failed login attempts. The recent activity log shows the system successfully identifying suspicious URLs and phishing emails.

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Deploy

 **Phishing Email Detector**

Email Subject

URGENT: Your Account Will Be Suspended!

Email Body

Dear Customer,  
  
We have detected suspicious activity on your account.  
Please verify your information immediately at:  
<http://secure-verify-login-xyz.com/account>  
  
Failure to verify within 24 hours will result in account suspension.  
  
Security Team

Analyze Email

Phishing Indicators: • Urgent action required • Suspicious links • Brand impersonation • Poor grammar • Excessive caps

 **Results**

Risk Score

85%

URLs Found

1

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Deploy

Analyze Email

 **Results**

Risk Score

85%

URLs Found

1

HIGH RISK

 **Reasons**

- Urgency word: urgent
- Urgency word: immediately
- Urgency word: verify
- Found 1 suspicious URL(s)
- Suspicious URL: <http://secure-verify-login-xyz.com/account>

## Phishing Email Detector Results

The Phishing Detector analyzes email content for common scam indicators. In this example, the system assigned an **85% risk score (HIGH RISK)** after detecting multiple red flags: urgency keywords ("urgent", "immediately"), a suspicious URL, and potential

brand spoofing attempts. The module provides users with a detailed breakdown of reasons, enabling informed decision-making about email safety.

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Deploy ⋮

 **Suspicious URL Analyzer**

Enter URL to analyze

http://secure-login-update-verify.xyz

Analyze URL

**URL Red Flags:** • HTTP instead of HTTPS • IP address in URL • Multiple subdomains • Suspicious keywords • URL shorteners

 **Results**

Risk Score

40%

Domain

secure-login-update-verify....

 **SUSPICIOUS**

 **Reasons**

- ⚠️ Uses HTTP (insecure)
- ⚠️ Suspicious keyword: secure

## Suspicious URL Analyzer Results

The URL Analyzer module scans web addresses for security risks. In this example, `http://secure-login-update-verify.xyz` received a **40% risk score** and was classified as **SUSPICIOUS**. Detection was based on the use of HTTP (insecure connection) and suspicious keywords in the domain, helping users identify potentially malicious links before clicking.

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Deploy

 Login Attack Monitor

Stats

Log Attempt

 Log a Login Attempt

IP Address

192.168.1.100

Username

user123

☐ Login Successful?

Log Attempt

Attempt logged successfully

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Deploy

 Login Attack Monitor

Stats

Log Attempt

 Login Statistics

Total Attempts

8

Failed

8

Successful

0

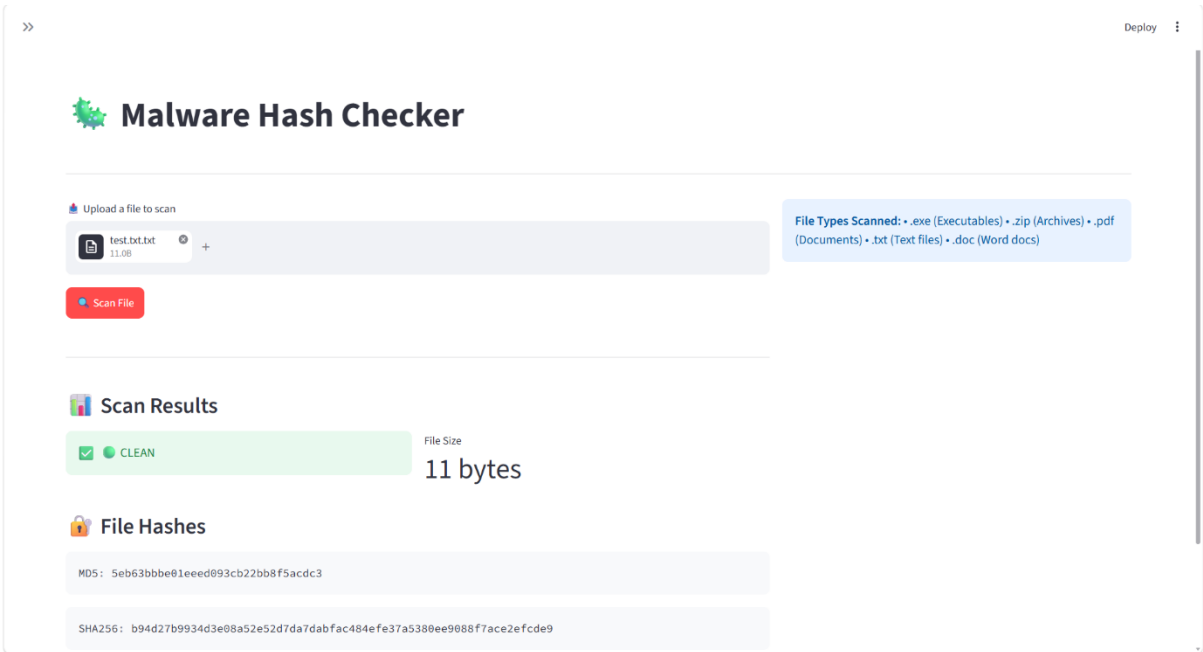
 Top Attackers

|   | ip            | attempts |
|---|---------------|----------|
| 0 | 192.168.1.100 | 8        |

## Login Attack Monitor - Brute Force Detection

The Login Attack Monitor tracks authentication attempts and identifies potential brute force attacks. The system records each login attempt with IP address, username, timestamp, and success status. When an IP exceeds **5 failed attempts within 5 minutes**, the module triggers an alert and automatically blocks the offending IP

address, preventing further unauthorized access attempts. This proactive security measure helps protect applications against credential stuffing and brute force attacks.



## Malware Hash Checker - File Security Analysis

The Malware Hash Checker module analyzes files by generating MD5 and SHA256 cryptographic hashes and comparing them against a database of known malware signatures. In this example, a test file of 11 bytes was scanned and verified as **CLEAN** with no malicious matches found. The system provides detailed hash information, enabling users to verify file integrity and detect potential threats before execution.